

Houssam Yassin

300 Forrestal Rd, 304 Sayre Hall, Princeton, NJ, 08540
 hyassin@princeton.edu HoussamYassin.com +1 380.710.5218

EDUCATION

PhD in Atmospheric and Oceanic Sciences (AOS) [Anticipated: July 2022] 2017-present
 Princeton University, Princeton, New Jersey
 Advisor: Dr. Stephen Griffies
 Thesis: Surface quasigeostrophic theory with variable stratification.

BSc with Honors: Mathematical Physics 2013-2017
 University of Alberta, Edmonton, Alberta

PUBLICATIONS

Submitted & In-preparation

Yassin, H. & Griffies, S. M., (in prep.): Seasonality in surface quasigeostrophic turbulence with variable stratification. [arXiv:2110.04242](https://arxiv.org/abs/2110.04242)

Published

Yassin, H. & Griffies, S. M., (accepted): On the discrete normal modes of quasigeostrophic theory. *Journal of Physical Oceanography*. [arXiv:2101.07432](https://arxiv.org/abs/2101.07432)

Yassin, H., (2021): Normal modes with boundary dynamics in geophysical fluids. *Journal of Mathematical Physics*, 64, 093102. [doi:10.1063/5.0048273](https://doi.org/10.1063/5.0048273)

Van den Bremer, T. S., **Yassin, H.** & Sutherland, B. R. (2019): Lagrangian transport by vertically confined internal wavepackets. *Journal of Fluid Mechanics*, 864, 348-380. [doi:10.1017/jfm.2019.30](https://doi.org/10.1017/jfm.2019.30)

AWARDS

Princeton Energy and Climate Scholar, Princeton University	2019-2022
Geophysical Fluid Dynamics Fellow, Woods Hole Oceanographic Institution	2019
Centennial Fellowship in the Natural Sciences, Princeton University	2017-2021
Dean's Silver Medal in Science, University of Alberta	2017
NSERC Undergraduate Student Research Award (x3)	2015-2017

OUTREACH & SERVICE

Co-presenter, Princeton Day School Energy and Climate Scholars	2021-2022
--	-----------

Student representative, AOS DEI Committee, Princeton University	2020-2021
Co-organizer, AOS student seminar, Princeton University	2019-2020
Co-organizer, AOS workshop on Machine Learning and Climate Modelling	2019

PRESENTATIONS

Yassin, H. & Griffies, S. M., (Nov 2021): A submesoscale k^{-2} kinetic energy spectrum from surface quasigeostrophic theory with variable stratification. *74th Annual Meeting of the APS Division of Fluid Dynamics*. [Accepted in-person talk]

Yassin, H. & Griffies, S. M., (Sep 2021): Seasonality in surface quasigeostrophic turbulence with variable stratification. *NYU Courant Center for Atmosphere Ocean Science Student Seminar*. [Virtual talk]

Yassin, H. & Griffies, S. M., (Feb 2021): Discrete normal mode decompositions in quasigeostrophic theory. *Ocean Model Working Group Meeting for the Community Earth System Model*. [Virtual talk]

Yassin, H. & Griffies, S. M., (Jan 2021): Are the baroclinic modes incomplete? On the normal modes of quasigeostrophic theory. 2021 *DRAKKER Annual Workshop*. [Virtual talk]

Yassin, H. (Nov 2020): Normal modes with boundary dynamics. *73rd Annual Meeting of the APS Division of Fluid Dynamics*. [Virtual recorded talk]

PROFESSIONAL MEMBERSHIPS

American Physical Society	Member since 2020
American Meteorological Society	Member since 2020
American Geophysical Union	Member since 2021

TECHNICAL SKILLS

Python: pyqg, dedalus, xarray, cython

Fortran: MOM6

C++

LANGUAGES

English (Native)

Arabic (Native)